

CLASS XII · LECTURE 13

Data Models, Keys and Relationships

Both options · Two periods · Section C: data models. Section B: primary and foreign key

By the end of this lecture you can

- Explain hierarchical, network and relational models.
- Define primary key and foreign key with one table example.
- Draw a one-to-many relationship.
- State 1NF, 2NF and 3NF in one sentence each.

A data model is the shape of the data**DEFINITION Data model**

A data model is a way of organising data and of showing how one piece is related to another. The model comes before the software. Access implements the relational model.

Model	Shape	What to say about it
Hierarchical	A tree. One parent, many children	Like folders, or a college containing classes containing students. Simple. Awkward when a child needs two parents, because the tree does not allow it
Network	Records linked as a graph. A child may have more than one parent	More flexible than a tree. Harder to design and to change. Not the model Access uses
Relational	Data in tables. A row is a record. A column is a field. Tables are linked by keys	The model in the paper and in MS Access. Questions are asked in SQL. This is the one to give the longest paragraph
Object-oriented	Data and the operations on it live together as objects	Mention it as a fourth model if the question says “all”. One sentence is enough

EXAM TIP The tree question

“The model that organises data in a tree” is hierarchical. “The model that allows queries through SQL” is relational. Those two multiple-choice answers are a pair. Do not swap them.

Keys

A **primary key** is a field, or a group of fields, that uniquely identifies each row. It cannot be empty, and no two rows may share it. Roll number in a Student table is a primary key. Name is a bad primary key because two students can share a name.

A **foreign key** is a field in one table that refers to the primary key of another table. It is how a relationship is stored. It may repeat, because many children can point at one parent. It should not point at a parent that does not exist.

DeptCode	DeptName
CS	Computer Science
PE	Pre-Engineering

Table DEPARTMENT. DeptCode is the primary key.

RollNo	Name	DeptCode
1101	Ali	CS
1102	Sara	CS
1103	Hina	PE

Table STUDENT. RollNo is the primary key. DeptCode is a foreign key pointing at DEPARTMENT.

This relationship is **one-to-many**: one department has many students. One student here belongs to one department. A **one-to-one** relationship is rarer: one person, one national identity record. A **many-to-many** relationship, such as students and subjects, is not stored as a single foreign key. It is broken into two one-to-many links through a third table, for example ENROLMENT(RollNo, SubjectCode).

ON THE BOARD Primary key and foreign key, six lines

A primary key uniquely identifies a row and cannot be null. Example: RollNo in STUDENT. A foreign key is a column that matches the primary key of another table. Example: DeptCode in STUDENT, which refers to DeptCode in DEPARTMENT. The primary key does not repeat. The foreign key may repeat. That repetition is the “many” side.

Normal forms, in case the question goes one step further

Normalisation is arranging tables so that facts are not repeated and updates stay sane. Three levels are enough.

Form	Rule in one sentence
1NF	Each cell holds a single value. There is no list inside a cell, and no repeating group of columns such as Subject1, Subject2, Subject3.
2NF	The table is in 1NF, and every field that is not part of the key depends on the whole key, not on only one part of a two-field key.
3NF	The table is in 2NF, and no non-key field depends on another non-key field. If Course determines Teacher, Teacher belongs in a Course table, not copied onto every student row.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. The model that organises data as a tree is:

- A) Relational
- B) Hierarchical
- C) Flat file only
- D) Spreadsheet

2. A key that uniquely identifies a record is the:

- A) Foreign key
- B) Primary key
- C) Caption
- D) Report

Short answers

S1. Differentiate primary key and foreign key.

S2. Define a one-to-many relationship and give an example.

Detailed answer

L1. What is a data model? Explain its types.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. B. One parent, many children, no second parent.

MCQ 2. B. The foreign key is allowed to repeat. It points at a primary key.

S1. A primary key uniquely identifies each row and cannot be null. RollNo in a Student table is a primary key. A foreign key is a field in one table that refers to the primary key of another table, such as DeptCode in Student referring to DeptCode in Department. A primary key does not repeat. A foreign key may repeat, which is how one department can have many students.

S2. In a one-to-many relationship, one row of the first table matches many rows of the second, and each row of the second matches only one row of the first. One department has many students. DeptCode is the primary key in Department and a foreign key in Student.

L1. A data model describes how data is organised and related. The hierarchical model is a tree: each child has one parent, for example a college containing classes containing students. It is simple and it cannot easily show a child with two parents. The network model allows a record to have more than one parent, which is more flexible and harder to maintain. The relational model stores data in tables made of rows and columns and links tables by keys. It is the model used by MS Access, and questions are asked with SQL. A primary key identifies a row; a foreign key refers to a primary key in another table. The object-oriented model keeps data and its operations together as objects. For this paper the relational model is the one used in practical work.